

DistilBERT fine-tuning Result

In this pipeline I didn't use any data cleaning or data augmentation I just concatenate query + image descriptions in one column name text , then renamed Toxic Category column to labels

I think our data doesn't need any data cleaning especially when we use model like BERT
Maybe i'm wrong maybe not but I just do this for experiment task .

First let me show selected Parameters & hyperparameters

LoRA :

r : Rank (Controls adapter size --> 16)

lora_alpha : Scaling factor for the LoRA output 32

Dropout : 0.2

LoRA

```
from peft import LoraConfig, get_peft_model, TaskType

peft_config = LoraConfig(
    task_type=TaskType.SEQ_CLS,
    r=16,
    lora_alpha=32,
    lora_dropout=0.2,
    target_modules=[
        "q_lin",
        "k_lin",
        "v_lin",
        "out_lin"
    ],
)

model = get_peft_model(base_model, peft_config)
model.print_trainable_parameters()
```

Train_args :

LR : from bert papers 1e-3 is good choice to start with

num_epochs : 3 is good to avoid overfitting

Train_args

```
training_args = TrainingArguments(
    output_dir="./lora_distilbert",
    learning_rate=1e-3,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=3,
    eval_strategy="epoch",
    logging_strategy="steps",
    logging_steps=50,
    save_strategy="epoch",
    load_best_model_at_end=True,
    fp16=torch.cuda.is_available(),
    metric_for_best_model="f1",
)
```

Result :

```
trainer.train()
```

wandb: **ERROR** Failed to detect the name of this notebook. You can set it manually with the WANDB_NOTEBOOK_NAME environment variable to enable code saving.
wandb: Currently logged in as: **medoeisa28** (**medoeisa28-alahein-international-university**) to <https://api.wandb.ai>. Use `wandb log --relogin` to force relogin

Tracking run with wandb version 0.19.11

Run data is saved locally in: `C:\Users\Mohamed\cellula-training\week-1\wandb\run-20250921_175333-rsji81gx`

Syncing run **stoic-sponge-4** to [Weights & Biases](https://wandb.ai) ([docs](#))

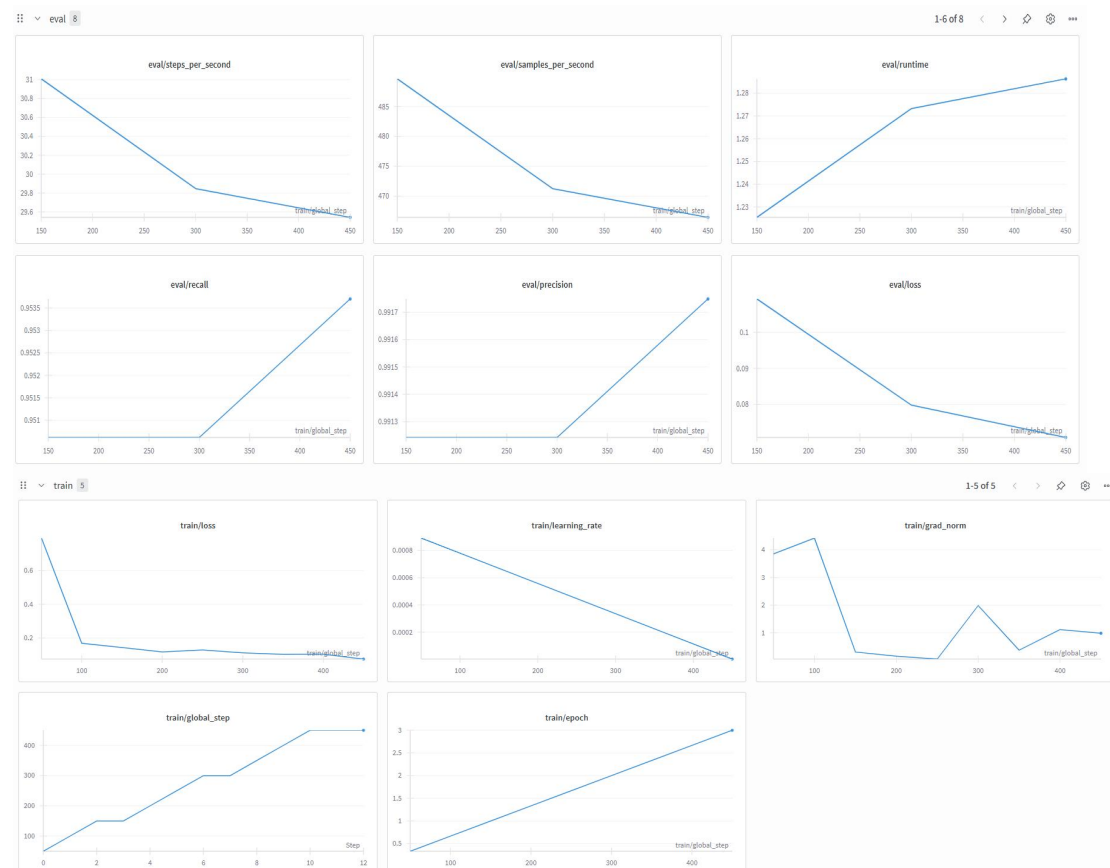
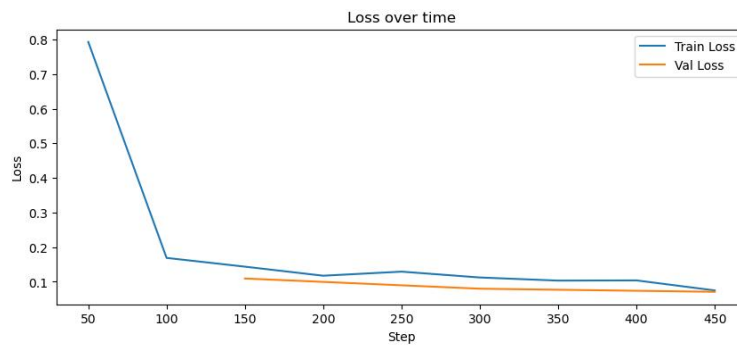
View project at <https://wandb.ai/medoeisa28-alahein-international-university/huggingface>

View run at <https://wandb.ai/medoeisa28-alahein-international-university/huggingface/runs/rsji81gx>

[450/450 00:43, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.143500	0.109283	0.973333	0.991242	0.950617	0.963696
2	0.112200	0.079784	0.973333	0.991242	0.950617	0.963696
3	0.074800	0.070797	0.975000	0.991749	0.953704	0.966476

TrainOutput(global_step=450, training_loss=0.1939519002702501, metrics={'train_runtime': 50.1954, 'train_samples_per_second': 143.439, 'train_steps_per_second': 8.965, 'total_flos': 490073164185600.0, 'train_loss': 0.1939519002702501, 'epoch': 3.0})



Notes :

The results show that the model has learned well without overfitting. However, because the dataset is too small to be reliably split into training, validation, and test sets, we cannot be certain that the model will perform equally well on unseen data. Applying data augmentation and balancing the classes would likely lead to significant improvements in performance, especially given the strong underlying model we are using.